

Statistical Inference for Model Checking of Uncertain Continuous-Time Markov Chains

Supplementary Information

Computation of the Probability Distributions Using EM and MCMC Algorithms

To approximate the behavior of the model under uncertainty, we must compute approximate probability distributions for model properties. In this case, the model properties are the results of 3 queries: the probability that the concentration of a given protein p is level i at time T , the expected percentage of a given protein p at time T , and the expected long-run percentage of a given protein p . For queries 1 and 2, we examine time $T=0$ through $T=10$ because results generally reach a steady state at $T=10$. We run each query on CTMC submodels that are partitioned from 3, 5, and 7 subintervals. Once query results are recorded, we calculate the mean and standard deviation for all the partitions for a single query. The mean and standard deviation act as the initial parameters fed to the EM and MCMC algorithms. The input of the EM algorithm is the mean, standard deviation, and observed data. Here, the observed data are the query results on the CTMC submodels. Using the current parameter estimates (mean and standard deviation), the expectation step creates a function for the expectation of the log-likelihood. The resulting function is passed to the maximization step, which computes parameters maximizing the expected log-likelihood. These parameter estimates are then used to determine the distribution of the latent variables in the next expectation step. The algorithm continues until convergence and returns the mean and standard deviation of the approximated probability distribution of our model's behavior.

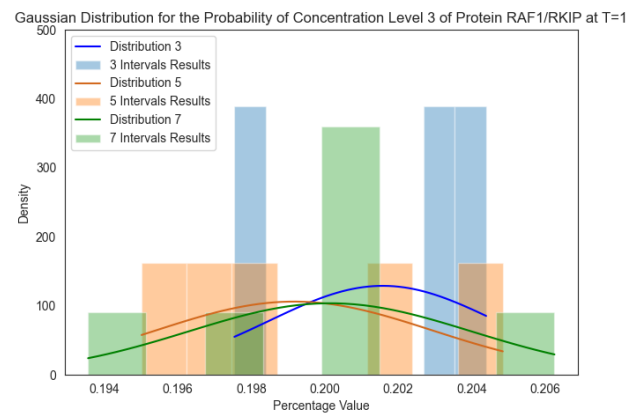
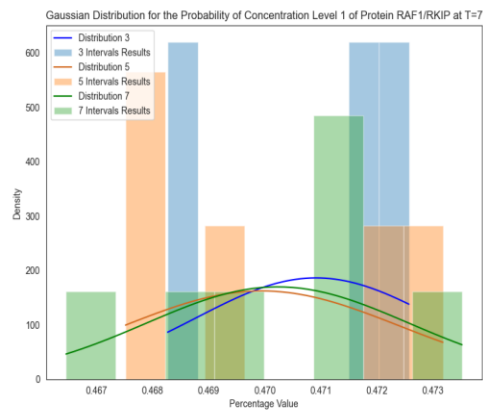
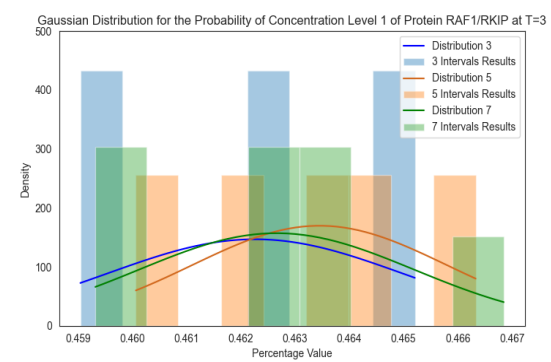
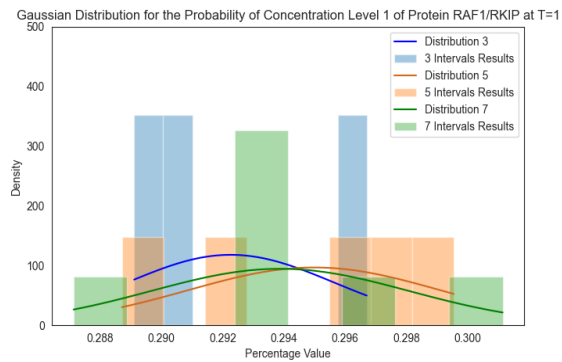
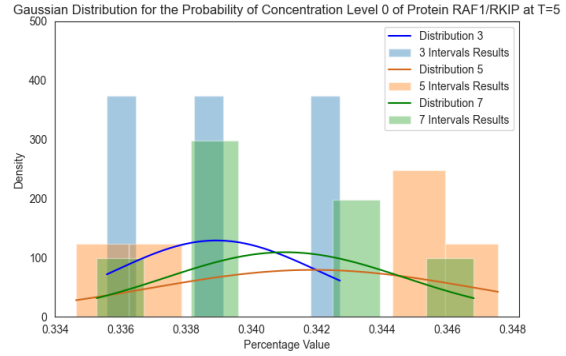
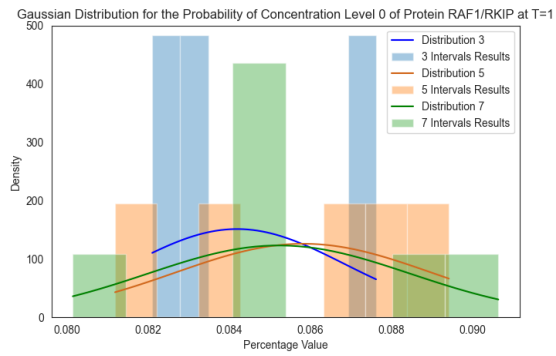
The MCMC algorithm is also initialized using the mean and standard deviation of each set of query results from the CTMC submodels. Given the probability distribution of query results, MCMC constructs a Markov chain whose elements' distribution approximates it. The goal is to have the resulting Markov chain's equilibrium distribution match the target distribution. Using the Metropolis–Hastings algorithm, the simulation takes some function proportional to our desired distribution. It is initialized at a random starting point to choose a starting proposal distribution. At each step, a new sample is proposed by picking from the proposal distribution. This sample is either rejected or accepted depending on the acceptance ratio. If accepted, the point is added to the sequence. The algorithm continues until convergence and returns the mean and standard deviation of the approximated probability distribution of our model's behavior.

Details of Results

Using the original RKIP inhibited ERK pathway modeled as a CTMC, we introduced uncertainty to construct an ICTMC, as defined in Table 1 of the main paper. From the ICTMC, we partitioned the transition intervals into three different sizes: 3, 5, and 7 disjoint intervals. Thus, we now have 3 ICTMCs: one with transition intervals partitioned into 3 disjoint subintervals, one partitioned into 5 subintervals, and another partitioned into 7 subintervals. From these ICTMCs, we construct multiple CTMC submodels using the methodology outlined in Sections 4 and 5 of the main paper. We ran 3 queries (as previously defined) on 2 different proteins: RAF1/RKIP and RAF1/RKIP/ERK-PP.

Results of each query can be found in Figures 3, 4 and 5 of the main paper. Looking at Figures 3 and 4, it is clear that not all submodels exhibit the same behavior. Protein concentration levels over time can vary greatly depending on the transition probabilities of the respective CTMC submodel. Figure 5 reports the long-run percentage of both proteins in every submodel. The long-run percentage of RAF1/RKIP is lower for models constructed from the lowest subintervals or lower transition probabilities. These models report a long-run percentage as low as about 22.5-25.5%. As the transition probabilities increase, there is generally an increasing trend in the long-run percentage of RAF1/RKIP. These models report a long-run percentage as high as about 24.5-26%. The opposite pattern is true for protein RAF1/RKIP/ERK-PP. The long run percentage of RAF1/RKIP/ERK-PP is higher for models constructed from the lowest subintervals or lower transition probabilities. These models report a long-run percentage as high as about 64-65%. As the transition probabilities increase, there is generally a decreasing trend in the long-run percentage of RAF1/RKIP/ERK-PP. These models report a long-run percentage as low as about 61.5-63%.

To gather submodel query results into final approximate probability distributions, we initialize the EM and MCMC algorithms using the mean and standard deviations of the query results. After running these algorithms, we have constructed approximate probability distributions for results from models constructed from 3, 5, and 7 disjoint subintervals. Results of the EM and MCMC algorithms can be seen in Figures 1, 2 and 3, but all results are listed in detail in “CTMC_All_Results_Stats.csv”. Reviewing these results, we see that the protein concentration level over time can vary greatly depending on the number of disjoint subintervals used to construct the CTMC submodels. The center of the distribution varies depending on the number of intervals. For RAF1/RKIP, the expected long-run percentage from 5 subintervals is *higher* than models constructed from 3 or 7 subintervals. However, for RAF1/RKIP/ERK-PP, the expected long-run percentage from 5 subintervals is *lower* than models constructed from 3 or 7 subintervals.



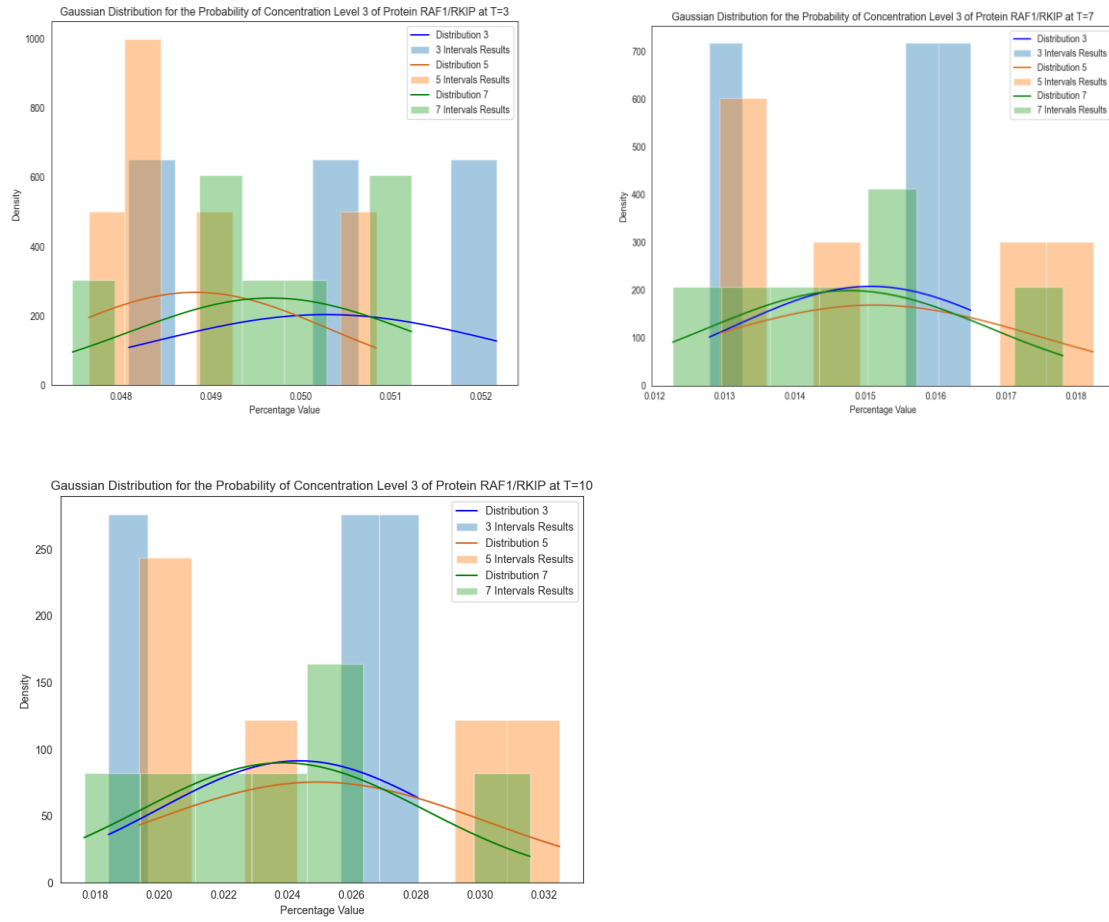
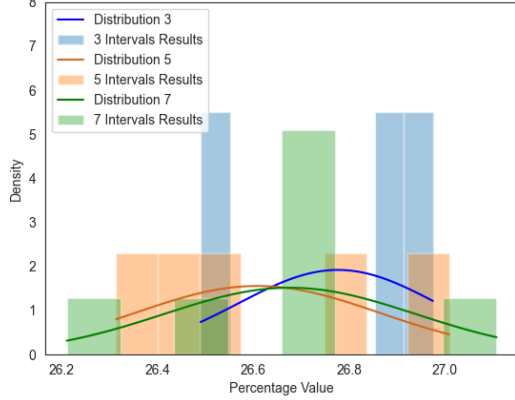
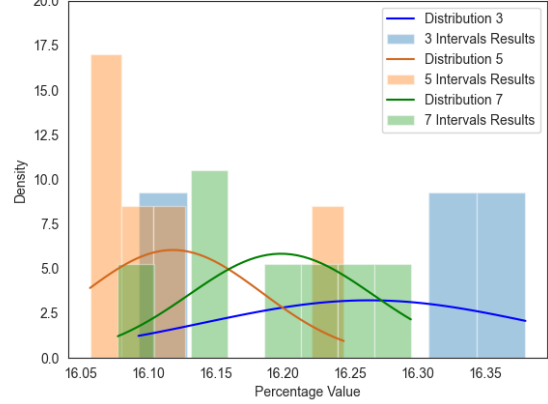


Figure 1: Plots reporting the approximate Query1 results on RAF1/RKIP using the Expectation Maximization (EM) algorithm.

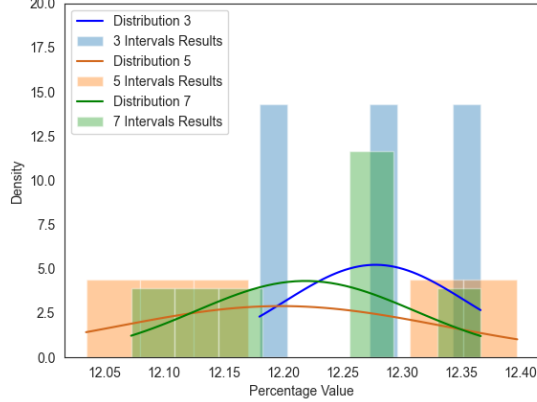
Gaussian Distribution for the Expected Percentage of Protein RAF1/RKIP at T=1



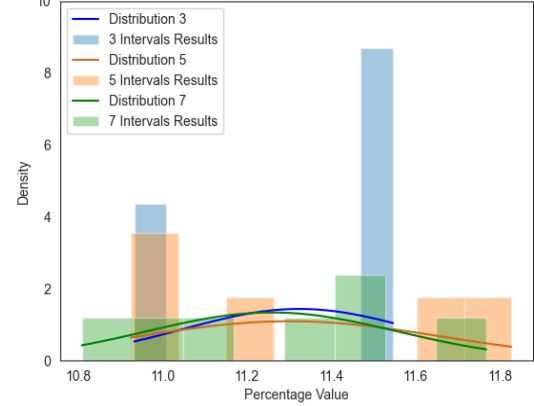
Gaussian Distribution for the Expected Percentage of Protein RAF1/RKIP at T=3



Gaussian Distribution for the Expected Percentage of Protein RAF1/RKIP at T=5



Gaussian Distribution for the Expected Percentage of Protein RAF1/RKIP at T=7



Gaussian Distribution for the Expected Percentage of Protein RAF1/RKIP at T=10

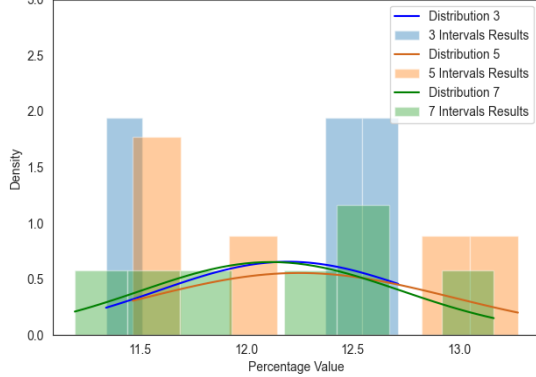


Figure 2: Plots reporting the approximate Query 2 results on RAF1/RKIP using the Expectation Maximization (EM) algorithm.

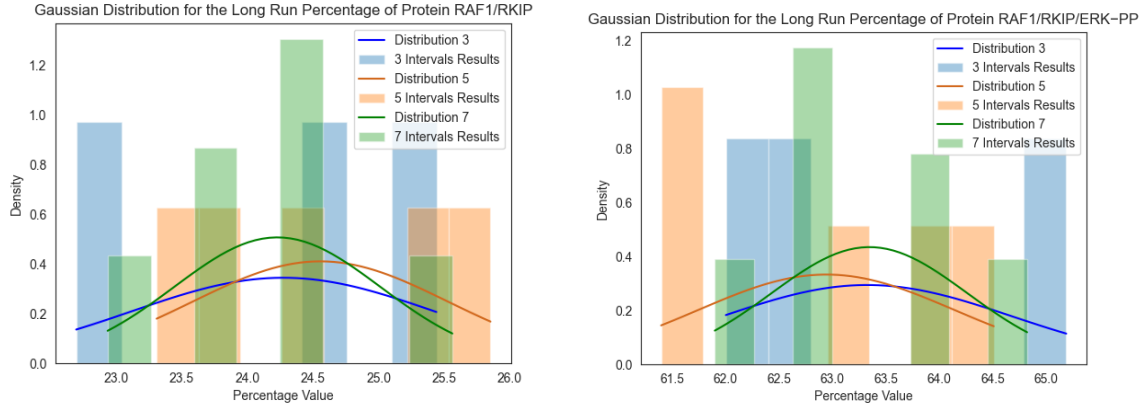


Figure 3: Plots reporting the approximate Query 3 results on both RAF1/RKIP and RAF1/RKIP/ERK-PP using the Expectation Maximization (EM) algorithm.

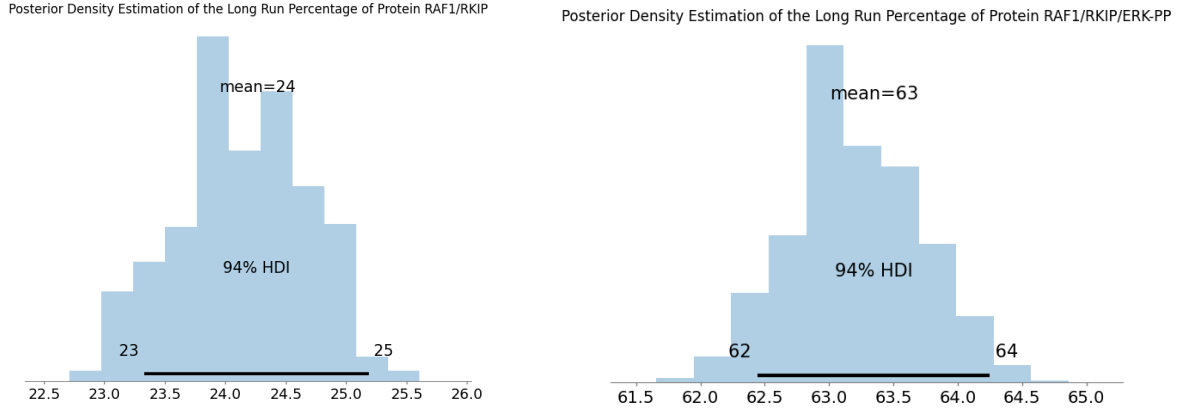


Figure 4: Plots reporting the approximate Query 3 results on both RAF1/RKIP and RAF1/RKIP/ERK-PP using the MCMC algorithms.

The approximated probability distributions produced by MCMC show comparable trends with the EM algorithm. Figure 4 displays approximate probability distributions for Query 3 for both proteins using a 3-subinterval construction. Again, the center of the distribution varies depending on the number of intervals. For both RAF1/RKIP and RAF1/RKIP/ERK-PP, the expected long-run percentage from 5 subintervals is higher than models constructed from 3 or 7 subintervals. To determine which submodels most closely resemble the original model, we compare the difference between the mean of the approximated probability distribution and the original query results on the original model. For practicality, the main paper only shows results at $i = 1$, $T = 5$ for Q1, and $T = 5$ for Query 2. All results can be found on Zenodo, but a summary of all results will be discussed here.

Query	Total	Method	3	5	7	Total % Below Average
Q1 on x	83	EM	12	15	15	69.167%
		MCMC	12	14	15	
Q2 on x	26	EM	5	4	4	86.667%
		MCMC	5	4	4	
Q3 on x	4	EM	1	0	1	66.667%
		MCMC	1	0	1	
Q1 on y	62	EM	10	11	10	51.667%
		MCMC	10	11	10	
Q2 on y	15	EM	2	2	3	50.000%
		MCMC	2	3	3	
Q3 on y	4	EM	1	0	1	66.667%
		MCMC	1	0	1	

Table 1: Report detailing the number of approximated query results that are defined as close to the query result on the original model. RAF1/RKIP is referred to as protein x , and RAF1/RKIP/ERK-PP is referred to as protein y .

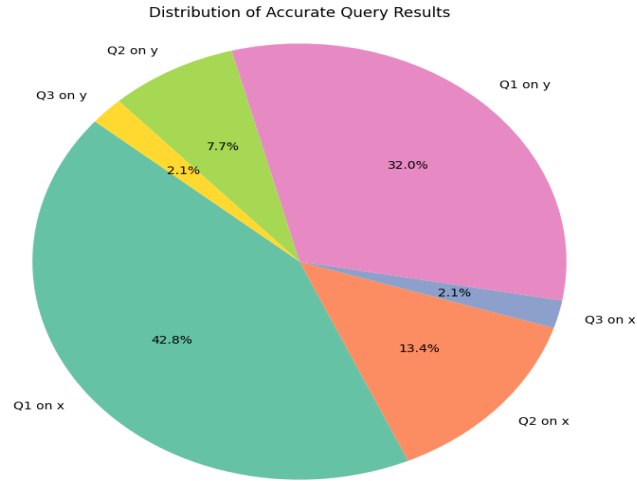


Figure 5: Distribution of query results that are defined as close to the query result on the original model.

Table 1 and Figure 5 detail the accuracy of each query using the EM and MCMC algorithms. In total, both EM and MCMC each accounted for 97 query results where the difference with the original query result was less than the mean difference for that query. Meaning, the EM algorithm produced 97 approximated probability distributions (across all queries and number of subintervals) whose difference from the original query results was less than the average

difference. 62 of these occurrences were from submodels constructed using 3 disjoint subintervals, 64 from submodels constructed using 5 subintervals, and 68 from submodels constructed using 7 subintervals. From these results, we conclude that the EM and MCMC algorithms produce results of comparable accuracy. There is no evidence to suggest that one algorithm may be better suited for approximating model behavior than the other. However, the number of subintervals may impact accuracy. As the number of subintervals increases, so does the number of results close to the original query. Although, it is important to note that only 3 different subinterval sizes were tested. To prove that accuracy increases with the number of subinterval partitions, more subinterval partitions must be tested.

Evidence does suggest that some queries are easier to approximate than others. Query 1, which calculates the probability that the concentration of a given protein p is level i at time T , is more easily approximated for RAF1/RKIP. 69.167% of the approximate query results are not far from the original model results. This is much larger than the 51.667% of approximate query results for RAF1/RKIP/ERK-PP. Of the 74.8% of accurate results covered by Query 1, RAF1/RKIP accounts for 42.8%, while RAF1/RKIP/ERK-PP accounts for only 32.0%. Similarly, Query 2, which calculates the expected percentage of a given protein p , at time T , is easier approximated for RAF1/RKIP. 86.667% of the approximate query results are not far from the original model results. This is much larger than the 50% of approximate query results for RAF1/RKIP/ERK-PP. Of the 21.1% of accurate results covered by Query 2, RAF1/RKIP accounts for 13.4%, while RAF1/RKIP/ERK-PP accounts for only 7.7%. Query 3, which calculates the expected long-run percentage of a given protein p , performs equally well on both proteins. Of the 4.2% of accurate results covered by Query 3, both proteins account for 2.1% each. A further biological analysis of the original model is needed to explain better performance on RAF1/RKIP queries over RAF1/RKIP/ERK-PP queries.